ENGLISH

Chapter 3

Stress Control

I, like everyone else, don't like the feeling of being stressed, but I have grown accustomed to the idea that life inevitably involves stress. Whether it was my high school classmates (most of whom were very driven), the athletes I've met, or the students, professors, and people in tech or creative fields around me, I have always found that stress accompanies ambition, and that for many of the top achievers, it's a constant. The weird thing is that many of the unambitious people I have known often said they were stressed out too. Lack of direction, lack of money, the feeling they were slipping behind in life, health issues, and more... Eventually I had to conclude that this was indeed part of life—like breathing, sleeping, or eating.

That meant there had to be a good way to address it. The problem was that when I looked to see what the science said about stress, it was uniformly negative. It made Clear that stress wasn't just uncomfortable, it was damaging to the brain and body. And that only made stress more stressful. Moreover, the whole framing of stress seemed wrong, or at least rather one-sided.

For instance, "fight or flight" is one of the most common phrases we hear when the topic of stress comes up. It's easy to remember and it seems science-y—because it sounds like it could describe how the brain and body respond to stress. Unfortunately, it's more misleading than informative. We also often hear that the stress response is an inconvenient holdover from our

evolutionary past. Meaning, the uncomfortable state of arousal we call stress was helpful for us long ago, propelling us to escape from sabertoothed tigers and other large predators, but it's no longer useful. I always appreciate when science shows up in a conversation, but I confess phrases like "fight or flight" make me shake my head. It's not just that they are misleading; they also don't provide any valuable information about how to navigate the stress response. And actually, the stress response is no holdover.

And research has shown that the beliefs we hold about stress can change how stress impacts our health, for better or for worse, so we need to be careful what we tell ourselves about itu Stress is an area I've thought a lot about from a neuroscience perspective. My lab has studied it in animals and humans. Decades into my career, I now know that there is, in fact, a good way to address it. There are excellent stress management tools based in science. So, given how pervasive stress is, I am especially excited to share the information in this chapter.

First let's begin with the facts. Forget sabertoothed tigers. Many of the same situations that stress us out now have been stressing out humans for thousands of years. An argument with a loved one, a persistent pain of unknown cause, a worry about a child, a looming payment, and on and on. Scenarios like these are not unique to modern times.

A good way to think about stress is this: Stress is nature's attempt to create a single solution to all matters that might be, or actually are, urgent. 1'11 say that again. Stress is nature's attempt to create

extreme, it can make people feel as if they have left their body, as if they are viewing themselves from above. While that all sounds dreadful, we must remember that the stress response is adaptive. The purpose of it is to put our brain and body into a state that gives us the best chance of addressing the problem at hand. A single solution. In an ideal world, the degree of stress response would perfectly match the stressor we face.

Running late? A little bit more arousal to help us focus on the key items we need to get out the door with grace would be perfect. See our toddler running toward the tramc? A huge surge in energy to quickly grab them is what we want! In fact, the parts of our nervous system that

govern stress—called the autonomic nervous system—can be calibrated to generally match our stress level to the stressor we face, from mild to severe, from brief to ongoing.4 But nowadays, the many comforts most people enjoy thanks to technology—indoor heating and air conditioning, abundant food and shelter, and all in all our general lack of actual survival stressors—have our autonomic nervous systems uncalibrated.

We are far more labile under the constant barrage of minor psychological stressors all day and night, and a lot of people are in a constant state of stress related to things they simply cannot control or influence even slightly. Put simply, the frequency and/or magnitude of the stress response is far too high relative to the stressors many people face. The occasional bout of stress lasting a few hours or less is no big deal, and even a few days is something our bodies can handle without long-term consequences. Once we experience a

few weeks to a month or more of daily (which usually also means nightly) stress, that's when we start to suffer. Chronically elevated resting blood pressure and heart rate, not sleeping well or enough due to rumination or feeling "tired but wired," and other shifts to our internal state become detrimental to our immediate and longterm health. We've all heard the laundry list of things that stress harms: our hormones, memory, mood, and decision-making ability, as well as the many diseases it makes worse or can bring about to our heart, brain, gut, and liver. The body of evidence that chronic stress is bad for us is vast and robust. Yet some stress is good for us.

In fact, the occasional spike in our stress hormones can increase testosterone5 and improve our immune system functioning during the hours that follow. In the short term, it can improve our memory. The problems arise when the intensity or the duration of the stress response is too great. Eliminating stress from our lives is unrealistic. So our goal with stress should be twofold.

First, we need to be able to adjust the intensity and the duration of our stress response so we can better handle stress in the moment. Second, many of us need to increase our stress threshold —that is, the subjective point at which events and thoughts trigger large arnounts of adrenaline release in our body—so our stress response is calibrated to any stressors we may encounter. Ahem, I mean, that we Will encounter. When adrenaline release occurs as the consequence of tiny things, that is not good. It negatively affects our physical health and relationships, including our relationship with ourselves.

Do you like being

around you when you're stressed? Didn't think so. We can change our relationship to stress through cognitive reframing, but an effective approach is to learn to quickly control our stress response once it hits, and to periodically reset our stress threshold to a level that best serves us. We can't always control what life throws at us, but we can indeed control our stress response, and we can raise our stress threshold. That's what this chapter is all about.

I first became interested in investigating the neuroscientific mechanisms behind stress back in 2015. At the time, grit, resilience, and mental toughness were popular themes in books and media. And those are great qualities. The examples—both historical and contemporary—of people being gritty and resilient and mentally tough are there to remind us that humans harbor within them an enormous capacity to manage and even channel their stress. While there is a lot known about the neural circuits and hormones that control the stress response, I specifically wanted to understand the mechanisms that determine its intensity and duration.

I wanted to know which features of the stress response, like heart rate, blood pressure, attentional narrowing, and so on, were amenable to adjustment in real time. I also wanted to know whether there are levers for adjusting the duration, the intensity, and maybe even the probability of the stress response getting triggered in the first place. I was able to find some data about stress inoculation,$ including some studies supporting the idea that certain forms of early-life stress can set the stage for resilience later in life by

hot baths, deep breaths, talk therapy. Great! Who doesn't love a massage at the end of a hard day? Meditation is powerful as a tool to buffer against the stress response; a randomized clinical trial conducted by researchers at Georgetown University found that people with generalized anxiety disorder who meditated experienced a noteworthy reduction in pro-inflammatory cytokines—chemical messengers essential to our immune system but that can also trigger a stress response—compared to those who didn't meditate.&? Other studies, including one done by my lab at Stanford in collaboration with my colleague David Spiegel's lab, also at Stanford, have shown Clear benefits of meditation practices as short as five minutes per day on stress, mood, and sleep.J-3 That might seem like a trivial level of time commitment, especially given the many benefits of meditation.

But I wanted to know how people can navigate stress while they are in the throes of the stress response. When their pulse is racing, their vision is blurry, their hands are shaking, and all they can think about is the stressor and the potential bad outcomes from it. That's when the capacity to manage stress matters most. Not offline, but in the moment. As I pointed out above, the stress response is universal.

Whatever the specific stressor might be, the stress response is the same. It varies in degree, not kind. This suggests that if we have reliable tools to reduce the stress response in one context, we ought to be able to reduce it in another. The problem? No one had dissected the universal features of stress control, only the stress response.

One valuable thing everyone should know about the stress response, though, is that it is asymmetric: It starts extremely fast and turns off relatively slowly. The adrenaline release that is the first domino of the stress response happens in seconds or less and can continue for several minutes after we perceive a stressor.J5 Think of a time when you were groggy from just waking up and you picked up your Phone to see a disturbing text. BOOM! You are Wide awake and ready to move in an instant. That's the speed of adrenaline in action.

Then, even after the stressor is resolved, the stress response persists. Imagine getting a text that says, "I think Mom was on the flight that just crashed," and ten minutes later, "I just spoke to her, she wasn't on there. She's fine." Your feeling of relief Will no doubt outpace how quickly your body settles down. That's an extreme example, but the same thing applies when someone almost sideswipes you while driving, and you swerve just in time to avoid a collision. A Shot of adrenaline wicks out through your body in an instant, and almost as fast, norepinephrine is released to ramp up attentional circuits in your brain.

Cognitively, you realize you're safe almost as quickly as you thought you were about to get hit, but your body and mind don't relax immediately. More often than not, we plummet into the stress response and it takes time to return to baseline. And that's when everything turns out Okay. When stressors persist, you experience wave after wave of adrenaline. And while the basic circuits for stress are the same in everyone, seldom discussed is that not everyone has the same threshold, or trigger point, for a stress response to kick off For some

people it's as if that first domino is fixed in cement. Such people are rare. For most people, however, the domino falls too easily. I wanted to understand this aspect of stress too, in order to develop tools anyone can use to raise their stress threshold, to allow them to calibrate their autonomic set point. That is, the balance point that our nervous system works to maintain—so that the stress response happens when it's needed and not when it isn't.

Even better would be to find ways such that when stress does happen, it occurs in proportion to the stressor. I wanted these tools to be mechanistic and behavioral, not pharmacological, because pharmacological tools —while they can be helpful—don't allow for realtime adjustments. You swallow a pill and that's that. And they have to be taken before stress arrives, which is problematic because we don't always know when stressors are coming. I also wanted to develop tools that do not rely on practice, like cognitive behavioral therapy, which is great but can take a fair amount of time to masterl and doesn't always translate across stressors very well; there is evidence both for and against generalizability.J-Z Like all protocols in this book, I wanted tools for stress management that could be effective the first time and every time because they are anchored in hardwired circuits and processes we all possess.

So I dove into the research on the physiology of stress to come up with a different way to frame the whole concept of stress. And stress control. That led to. The Seesaw Model Think about the stress response as existing somewhere along a spectrum of arousal, with

in the daytime and asleep for a major portion of the night—then your SNS activity levels Will dominate in the daylight hours and your PNS Will dominate at night. In fact, the incredible impact your circadian rhythm has on your ability to reason,2 exert physical effort,2-W and resist infections22 stems from how it controls the balance of SNS versus PNS activity. In the morning and early daytime, elevated SNS activity (sometimes called "sympathetic tone") caused by a natural and healthy increase in our baseline levels of cortisol makes us physically stronger and better able to focus, whereas in the later evening and at night, when our baseline cortisol levels are markedly lower, our parasympathetic tone is elevated, and we become mentally and physically weaker, and our immune system becomes far less capable of warding off infections. (Circadian variation in the expression of specific genes and the actions of various molecules— neuromodulators, hormones, peptides, etc.—are what underlie those dramatic morning-versusnighttime variations in autonomic tone, as discussed in chapter 1 The second category of things that control our autonomic nervous system, and thus our level of arousal, are the sensory inputs—or stimuli, as neuroscientists call them—that we experience. For example, our level of physical comfort or pain, how well-fed or hungry we happen to be (not surprisingly, hunger makes us more alert), whether we are in a brightly lit or dim environment, the environment is quiet or loud, et cetera. And, of course, by the presence or absence of psychological stressors.

In general, when we are hungry, thirsty, tired, or otherwise in need

of basic sustenance, our arousal levels increase and, with that, our propensity to think and move increases. It should be obvious Why that is adaptive. (It helps us seek out food, water, etc.) The third category of things that control our autonomic nervous system is pharmacologic. Caffeine and other stimulants are sympathomimetic, meaning they stimulate the SNS and increase arousal.24 Other drugs promote PNS activation, and thus decrease arousal levels. For example, alcohol and barbiturates indirectly increase PNS circuit activity,2 as does the molecule apigenin, which is found in chamomile tea" By the way, ifyou're wondering Why

alcohol initially makes people more talkative and alert before it makes them drowsy or has them pass out, it's because before it reduces central nervous system activity and lowers arousal, alcohol first transiently stimulates the SNS. "It lifts you up before it drops you," as they say.

Autonomic means "automatic" and indeed our sympathetic and parasympathetic nervous systems are, for the most part, not under our direct conscious control. We of course can adjust them indirectly. We just discussed three categories of ways to do that. The pathways and mechanisms by which different activities like exercise, or mechanical inputs like massage or a warm bath,2-g or drugs like caffeine22 elevate or lower arousal by adjusting the activity of our SNS or PNS are fairly well known, but again, it's crucial to note these are all indirect ways of adjusting autonomic arousal. As such, they tend to be Slow and hard to titrate—meaning hard to get just right.

Moreover, some, like barbiturates or

prescription stimulants, have addictive potential and/or are costly. There is, however, one major exception to all of this: one lever by which we can quickly, directly, and reliably change our level of autonomic arousal by shifting the balance of SNS versus PNS activity. That lever is our respiration pattern, aka our breathing. Why? Because the neural circuits that control our breathing are part of our autonomic nervous system.

In fact, they are the one (and only!) component of it that is both autonomic— breathing works just fine without us having to think about it—but that we can also consciously and directly control at any moment we choose. The same cannot be said about our heart rate, digestion, or immune system. As my colleague at Stanford, David Spiegel, MD, has described it to me, breathing and the neural centers that control breathing represent a bridge between our conscious and unconscious mind. We Will return to this idea shortly, but to understand the specific patterns of breathing that allow us to consciously and directly control our stress response, and to understand the best tools for adjusting our stress threshold, we need to take the autonomic spectrum—that continuum spanning from deep states of calm to intense states of panic I described earlier—and turn it into a seesaw. A seesaw?

Correct. The simplest model for understanding how to control stress is a seesaw: parasympathetic at one end and sympathetic at the other. When the seesaw is tilted toward the parasympathetic end, we feel calm and relaxed. Tilt it further in that direction and we start drifting off to sleep. A flat, balanced seesaw

limits the range of solutions you can come up with to a problem.3-Z It even changes your ability to link previous experiences in insightful ways.33 Most of us are familiar with this stressinduced black-and-white thinking. These facts should help alleviate some of the frustration you may experience when someone tells you to "just relax," and you can't. Or when someone says, "Don't worry, this too Shall pass." The first form of encouragement ignores that we must actively do something to our physiology to relax. You can't out-think adrenaline. The latter encouragement is spot on, as stress does indeed hinder our ability to see the big picture, but it's not particularly helpful in the moment.

It's sort oflike telling you that the splitting headache you're suffering from right now won't suck in the future. Fortunately, there is a way to get right to the physiology of the stress response. Rather than try to directly reduce sympathetic tone, you need to instead tilt the seesaw back toward the parasympathetic side by increasing the activity of your PNS. That is

going to be the fastest way to calm down in real time whenever you are feeling stressed. Resilience defined If you think about it, this is a big part of what resilience is really about.

It's not our ability to tolerate or eliminate the stress response but rather our ability to reduce its intensity quickly and in real time, when our arousal exceeds a useful level, so that we can move forward. Resilience is our ability to control stress. Sometimes that means going from extreme stress to very alert and energized, so we can perform a task and quickly problem-solve under intense

pressure. Other times it means taking ourselves from endlessly looping thoughts to deep sleep. Resilience is also about an ability to deliberately increase our stress threshold, so life's minor stressors don't disproportionally hijack our mind and body. Once you learn how to do those two things, then you Will truly be in the driver's seat. Maybe that all sounds overly simplistic.

Easy to say but harder to do. Well, it turns out there are science-supported tools to adjust your stress response in real time and to raise your stress threshold, and they work exceedingly well—and fast. They can also work the first time, and every time, because they are grounded in biological mechanisms and circuits that you already have inside you and can access. Some involve adopting a specific pattern of breathing, but they are not breathwork. After all, when we find ourselves in a high stress situation, we can't run off and do five minutes of box breathing!

Other protocols don't involve breathing at all. I've been blessed to play a partial role in the discovery and application of several of these stress control protocols, although like so many other scientific "discoveries," aspects of them were known about and used by nonscientists in other contexts first. None of us gets the patent. The following pages explain what they are, how they work, and how to use them.

STRESS CONTROL PROTOCOL Do One to Three Physiological Sighs to Reduce Stress Fast

WHAT TO DO

• Breathe in the pattern known as the physiological sigh. Inhale deeply through your nose until your lungs are full (your belly should expand out while you inhale), then sharply inhale a second time, also through your nose, to maximally inflate your lungs. Then slowly exhale through your mouth until your lungs are empty of air. Repeat this in-in-out breathing pattern up to three times, in succession, as needed to reduce your stress level (meaning, your level of autonomic arousal) fast.

HOW IT WORKS

Physiological sighs are, to my knowledge, the fastest and most effective way there is to deliberately calm yourself down. Often just one physiological sigh does the trick. That's because our state of mind and body changes our breathing but our breathing also changes our state of mind and body.3.4 Remember the locus coeruleus that releases norepinephrine to make us alert? Well, specific populations of

exhale. We need both oxygen and carbon dioxide, but we need them in the right balance. If we breathe too little, carbon dioxide builds up in our brain and bloodstream, leading to headaches and tiredness.3-$ If we breathe too much—rapidly inhaling and exhaling, or what we typically call hyperventilation—we expel too much carbon dioxide, which is problematic because carbon dioxide is crucial for making sure oxygen gets to our brain cells. If you've ever wondered Why

it's advised to breathe into a paper bag if you're having a panic attack, it's to recapture some of the carbon dioxide that would otherwise be expelled. In this chapter you'll learn faster and better ways to calm down, but the logic is the same.

Overbreathing causes anxiety, dizziness, confusion, poor sleep, and even muscle spasms. As mentioned in the introduction to this chapter, the neurons in our brainstem that control our breathing represent a bridge between our conscious and unconscious mind. They are equipped to sense the ratio of oxygen to carbon dioxide in our blood, which prompts any necessary alterations in our breathing pattern to keep us alive and well. That happens regardless of whether we are exercising hard or deeply relaxed —it even happens while we sleep.40 Unlike other autonomic neural circuitry, however, we can take immediate conscious control of these neurons. We can decide "I'm going to extend my next exhale a bit" or "I'm going to take one more inhale before holding my breath" and—voila!—we can make that happen.

We just don't have that level of conscious control over any other autonomic function. There are some reported examples of people who used biofeedback to learn to directly

control their heart rate or the size of the pupils of their eyes, but that requires a fair amount of training.4J Extended exhales and physiological sighs are something that anyone can do. Certain internal states are associated with certain respiratory patterns. And by deliberately changing how we breathe, we can shift our internal state. That's a big deal and something I wish they had taught us in grade school. Alas, it wasn't until recently that Western science embraced controlled studies of respiration and, in doing so, revealed what had been known in Eastern traditions for thousands of years.

We owe it to a small set of dedicated researchers, mostly neuroscientist Jack Feldman's lab at UCLA, for discovering how the brain controls breathing, both our unconscious reflexive breathing and deliberate breathing, and how they impact our emotional and cognitive states.4-Z Before and during our discussion on the Huberman Lab podcast, Jack told me stories of setting out to study this topic in the early 2000s, and a luminary of neuroscience making fun of him at meetings. He would jab: "Why would you study something as boring as breathing? Why don't you study something important, like (That guy studied memory.) Jack knew to ignore him and follow his intuition. His work ignited a field that is now positively impacting the mental and physical health of people of all ages. Detecting and modifying certain breathing patterns also turns out to be important for predicting heart attacks,44 strokes,45 and seizures,46 and for preventing sudden infant death syndrome (SIDS).4Z Some of his recent work has also shed light on the fact that some opioid drugs can lead

to death because of how they change the firing patterns of the neurons that control breathing. It is fair to say that Jack's work has literally saved lives. Breathing isn't just about getting oxygen into your lungs; it also helps control heart rate. While you can't directly control your heart rate, you can leverage the fact above and use your breathing to control it indirectly. Deliberately inhaling more vigorously or longer (or both) increases your heart rate, whereas deliberately exhaling more vigorously or for longer slows your heart rate down, through a process called respiratory sinus arrhythmia.

Exhales also activate a specific branch of the vagus nerve, which is part of the PNS, which in turn reduces the rate of electrical activity in neurons of the sinoatrial node of the heart, your heart's natural pacemaker. So deliberately emphasizing the intensity and/ or duration of your exhales for a few breathing cycles Will caaaaalm youuuu dooooown. The breathing pattern described in this protocol does just that. Once you learn the relationship between breathing

and heart rate, and deploy the physiological sigh to reduce your heart rate on the spot, it gives you a significant feeling of confidence in dealing with stress, and uncertain circumstances too. You no longer have to be afraid you Will get stuck in an anxiety response in a key moment, or any moment for that matter.

You also realize that what most people say to themselves or others when stress hits—"Take a deep breath"—is the exact opposite of what we need to do! When stress hits we should do a big exhale. It is when

Thus, physiological sighs, unlike just a big exhale or box breathing, are a three-pronged tool to quickly (1) Slow your heart rate, (2) reestablish proper oxygen—to—carbon—dioxide ratios in your brain and bloodstream, and (3) engage brain areas that let you think logically about your situation. It's adaptive biological engineering at its best and works to calm you down. Infants do physiological sighs reflexively. Animals do too.5J Observe a dog right before it goes down for a nap. You'll see that double inhale, long exhale pattern.

Rats physiological sigh when 5.2 placed into a claustrophobic environment. People reflexively do physiological sighs about once every few minutes when working5.3 and even in their sleep to reset their oxygen—to— carbon-dioxide ratio. In fact, you have dedicated sets of neurons in your brainstem for generating physiological sighs.55 Nature just doesn't create discrete neural pathways by accident; that's how important this breathing pattern is. One respiratory region, the pre-Bôtzinger complex, controls your rhythmic breathing, the regular alternation of inhales and exhales.5é But a second region, the parafacial nucleus, controls breathing that involves double inhales, double exhales, or any combination of inhales and exhales in ratios other than the 1:1 inhale-exhale pattern.5Z The parafacial nucleus does things other than generate physiological sighs but that's one of its primary functions. I'd like to clarify that physiological sighs are not breathwork.

Breathwork refers to setting aside five to twenty minutes to do a specific breathing pattern (or designated sequence of patterns) to get you into a specific state of mind and body.

Physiological sighs are a naturally occurring breathing cycle that we can deliberately recreate. And they are the fastest way to consciously tilt your seesaw to the parasympathetic side of your autonomic nervous system. Do it before public speaking or a hard conversation. Do it when you're stuck in trafic or late to an important meeting and starting to feel your tension level rise. Do it to stop yourself from overreacting to an offhand comment.

It won't guarantee that you'll always do the right thing in response, but it Will put you in a better frame of body and mind, so that you have the best chance of navigating the stress that comes from these or similar situations. There are data showing that doing physiological sighs outside of stressful moments 5.8 can help buffer you against future stress too. Several years ago, my lab collaborated with psychiatrist David Spiegel to investigate just that.5.2 We conducted a study that randomized 108 participants to one of four groups. One group did five minutes per day of cyclic physiological sighing: Inhale, inhale, long exhale, repeat, and so on for five minutes. The second group did five minutes per day of box breathing, a breathing technique I've mentioned before, which follows a pattern of inhale-hold-exhale-hold-repeat, where each inhalation and exhalation, and the holds, are the same duration, until five minutes have elapsed.

The third group did five minutes per day of cyclic hyperventilation: You take a big deep breath in through the nose and a shorter exhale through the mouth, putting more vigor into the

inhalation than the exhalation. A fourth group did five minutes per day of meditation, seated

with eyes closed, and focusing their attention on their breath but with no specific instructions about breathing pattern. Most participants, regardless of their group, reported an improvement in mood and outlook. But cyclic physiological sighing had a notably bigger effect. We also observed improvements in resting heart rate, improved sleep, and other metrics of well-being related to autonomic function. In other words, doing one to three physiological sighs is a great tool for when stress hits and doing five minutes of physiological sighs on repeat can help buffer us further against the inevitable stress of life.

Given what you now know about exhales slowing heart rate, oxygen and carbon dioxide levels, and the autonomic seesaw, it should make sense as to Why this works. Your brain and body were literally built to do this.

STRESS CONTROL PROTOCOL Use Deliberate Cold Exposure to Raise Your Stress Threshold

WHAT TO DO

• Put yourself in uncomfortably cold water three to seven times per week. This can be as simple as a cold shower, or as involved as an ice bath. For the duration, choose one ofthese options: • Timed protocol: One to three minutes per session. • Count "walls of adrenaline" protocol: Rather than committing to a specific amount of time, try committing to a specific number of mental "walls" to climb over. When you feel the impulse to avoid or exit the cold but you stay, that's one wall you've scaled. Decide in advance how many times you Will encounter this impulse and keep going.

You do not have to always increase the number of walls. There are days when you Will decide to "go over" five walls and other days where two walls Will be enough. • The water should be uncomfortably cold. This means it should immediately trigger your

the strength and hypertrophy adaptations that would otherwise occur. (Chapter 2 explains more about those adaptations, and how to achieve them.) That inhibition of muscle growth is likely due to the vasoconstriction the cold causes, which limits both nutrient delivery and waste removal from the recently trained muscles and also blunts protein synthesis.60 Wait at least Six hours after resistance training to do deliberate cold exposure, or do it before you train, or just save it for another day. Or do it any time relative to your resistance training, but be sure to end with a hot shower or bath to reestablish perfusion of the muscles to correct for this. One study argued that deliberate cold exposure done in the first Six hours after resistance training does not prevent strength adaptations, but most of the work on this issue indicates that deliberate cold exposure done in the Six hours or so after resistance training Will significantly reduce any hypertrophy or strength increases that would otherwise occur. 1 So it is best to simply do it before resistance training, or on rest days. Or allow Six or more hours to elapse after resistance training before deliberate cold exposure. Another common myth is that deliberate cold exposure is harmful for people wishing to limit their cortisol levels.

You often hear that people should avoid deliberate cold exposure to keep their cortisol levels low too. Yet the cortisol increases caused by deliberate cold exposure, while significant, are transient and small compared to the dramatic increase that a moderate intensity exercise session triggers. You won't hear that people should avoid exercise to keep cortisol low because the cortisol increase

caused by exercise is, again, transient and, what's more, beneficial. (Indeed, it is a key aspect of the specific and general adaptation triggers we discussed in chapter 2). Many of the "cold exposure is bad for you" stories involve people going into the cold for very long periods of time at excruciatingly cold temperatures, or combining deliberate cold exposure with hyperventilation breathwork and/ or jumping or diving under ice. All of which are dangerous and should be avoided. Now that those points are out of the way, let's look at Why deliberate cold exposure is beneficial. Done correctly, deliberate cold exposure is a powerful, safe, and reliable tool for rapidly adjusting your internal state and stress threshold.

There are two key phases, each with distinct benefits. First, the state shift that occurs during the cold exposure itself. Second, the state shift that occurs for many hours afterward. Anyone who has done deliberate cold exposure consistently—that is, three to seven times per week—will tell you there is a third phase too, which is the change in your subjective experience of stress and your ability to navigate other forms of stress. When you enter the cold, it sharply tilts your autonomic nervous system toward the sympathetic end of the seesaw.

It does this the first time and every time, and it happens fast, within one to two seconds upon entering the cold. Then, once you get out of the cold, you tend to remain in a very "alert but calm" state—meaning (in the context of the model I described) with an evenly balanced autonomic seesaw—for many hours afterward.

Deliberate cold exposure increases the levels of circulating catecholamines in your brain and body: adrenaline, norepinephrine, and dopamine, and increases in the hormone cortisol too. Some, including dopamine in particular, remain elevated long after you dry off and warm up. These catecholamines increase mood, energy, and alertness. They also can increase your stress threshold; more on that below.65 But first I want to pause for a moment because you may be asking this: Why would I want to trigger the release of these "stress" chemicals? Won't that make me feel more, not less, stressed?

Yes, but that's what this protocol is all about. We need to challenge the notion that stress is always bad for us. In fact, when we have control over when, how much, and especially for what purpose we experience stress, the impact of that stress can switch from being health-perturbing to healthpromoting, and this reshapes the body's response to all forms of stress.6é This is not "mind over matter," it's leveraging the brain circuits that convert the arousal response we call stress into something called eustress a term coined by Swiss physiologist Hans Selye.6Z (You may remember him from the previous chapter.) He was nominated for the Nobel Prize seventeen times for his work on this and related topics, long before we knew what the underlying mechanisms were, but as those mechanisms have become elucidated, many of his theories have largely been deemed correct. We now know that your prefrontal cortex —that patch of brain real estate involved in context-dependent decision-making (as well as

have been in before, but what we believe to be true about stress—affects how we respond to it. Work from Alia Crum's lab at Stanford shows that humans who learn about some of the (true) positive effects of stress-related arousal on physical health and mental performance experience those positive effects when they are subjected to stress" Her lab also showed that if people learn a bunch of (also true) information about how stress can make us physically unhealthy and cognitively impaired, well, that's what happens to them when they get We can thank our prefrontal cortex for that ability to reframe stress in ways that change its actual effects on us. It's not just a decrease in its negative effect but an inversion from deleterious to

beneficial. There's more that shapes our experience of stress as good or bad. Perhaps one of the most interesting things that determines the outcome is whether the stressor has a known end point and resolution that we control.Z2 With that in mind, it should now make perfect sense Why deliberate cold exposure meets all the criteria of eustress, and Why I coined it deliberate.

If you are the one who elected it, and you control its duration, and you understand how it is good for you, you can turn stress into eustress. So how, exactly, is it good for you? Two ways. First, the stress of deliberate cold exposure helps us become better at handling other moments of stress. That's sometimes called cross adaptation.

It's often studied in the context of physical stress, where exposure to one type of stress—say, exercising in extreme heat—prepares a person to tolerate another type of stress, such as thriving in environments with low oxygen levels.Z3 The idea

is that we can precondition ourselves to cope with challenges by seeking out situations that draw on similar aspects of our physiology. Cold water swimming has been proposed as a way to improve surgical outcomes,Z4 for example, because the stress of the former helps patients better handle the stress of the latter. Controlled studies showing that cross adaptation focused on physical stress can prepare us to face emotional stress are slowly accruing. In one study of eighty-six people, combining cold exposure with breathing techniques changed the participants' perception of stress in their everyday lives.Z5 Participants rated statements such as "You feel calm" and "You enjoy yourself" following a few minutes of vigorous breathwork and a quick cold shower every morning for a period of two weeks. And participants who did both during that period reported a greater sense of control over irritations, and stronger feelings that life was going their way and that they were on top of things, compared to participants who did only the breathwork or the cold exposure, or neither.

Some data are more tangential in nature. For example, cold exposure appears to improve the body's response over time to hypoxia, or low oxygen levels in the blood, which is relevant because hypoxia can rapidly set in when we hit a state of anxiety or panic.7 Some preliminary evidence has pointed to a possible role for cold exposure in the treatment of depression, not as a replacement for medication but as an accompaniment.ZZ In 2018, a group of researchers from the UK reported that a twentyfour-year-old woman who had been treated for

major depressive disorder and anxiety since the age of seventeen with little to no relief began weekly sessions of cold water swimming in an attempt to find something that could diminish, or negate, the need for medication after she became a mother. The intervention worked for her. Within four months, she no longer required drug treatment.Z8 That was a case study of just one person, but interesting nonetheless. A 2023 review of fourteen studies on the mental health benefits of outdoor cold-water swimming found this practice to be associated with stress reduction, mood improvement, and a positive outlook.Z2 That result is not necessarily entirely tied to cold; being outdoors and part of a community (many people do winter swims in groups or with friends) certainly contributes, but cold exposure is responsible for at least some of this benefit. Other work has also shown a connection between winter swimming and well-beingg-Q In one study, fifty-three people in the UK who participated in eight weekly sea swims reported a reduction in both depression and anxiety.8-J Interestingly, "confronting a challenge" was commonly reported as a reason Why the experience was rewarding.

In another investigation, forty-two undergraduates at the University of Chichester, in the UK, completed a mood profile questionnaire before and after cold immersion, with an additional

twenty-two students completing the questionnaire twice but without cold immersion in between. 2 Energy and self-esteem increased significantly in the experimental group, which also experienced a decrease in tension, anger, and depression. No

plunge or cold shower more than getting into it. In all seriousness, very few people actually enjoy deliberate cold exposure while they are in the cold. It is a potent stressor and no amount of learning about eustress Will change that. In fact, I often find myself cursing Selye when I'm in the cold. But once we get out, ahhh, that's when the real magic starts.

The Sharp increase in adrenaline, norepinephrine, and cortisol we just triggered also sets in motion a dramatic and prolonged increase in dopamine—the molecule that, as we Will discuss more in chapter 6, induces a heightened sense of motivation and focus and, with the pain of the cold now gone, buzz. Dopamine doesn't act alone to create the postdeliberate-cold state, but it is a central player in that neuroendocrine cocktail. It's the perfect state for a work session, for exercising, socializing, pretty much any activity that requires us to be alert and focused. If the way I'm describing the post-cold state sounds almost drug-like, it is. Only better.

One study found that full-body (up to the neck) immersion in water at about 57 degrees Fahrenheit for one hour triggered a more than fivefold increase in norepinephrine levels compared to baselineg-6 It also produced a rise in dopamine levels in the body, raising them more than twofold for hours afterward. We call this peripheral dopamine, because it was measured from the body. We don't yet know the exact relationship between dopamine levels in the body and in the brain, but we do know they tend to increase and decrease in parallel. And there is emerging evidence that when peripheral dopamine is elevated, it increases dopamine in

reward and motivation circuits in the brain in a dose-dependent manner.8 When adrenaline is released in the body, it doesn't cross the bloodbrain barrier but it does activate receptors in the vagus nerve that project from the body to the brain. In doing so, it causes the locus coeruleus to release norepinephrine, which increases alertness in the brain. So there is reason to assume that peripheral increases in catecholamines triggered by deliberate cold exposure are mirrored by similar increases in the brain. It is worth noting that drugs known to sharply increase dopamine and epinephrine, such as amphetamines or cocaine, also produce fast, large increases in catecholamines but with the important caveat that they cause levels of the catecholamines to rise so fast and then drop so fast, that they plummet below pre-drug levels. That is part of what, as Anna Lembke, MD, explained in Dopamine Nation, puts people in a state of psychological and physical discomfort, and leaves them craving more of those drugs right away, which partially explains Why they are addictive.

Deliberate cold exposure then seems an almost perfect way to significantly increase and maintain elevated levels of catecholamines. You get the mental focus and mood benefits they provide, and you also raise your stress threshold. Most of us don't have an hour available to spend sitting in chilly water, however. Fortunately, you can use colder temperatures for much shorter periods of time to get the same effect. Studies show that even thirty seconds can workg-2 Yet I suggest one to three minutes or two to ten "walls." As mentioned

disruption in its communication with subcortical areas. This is true of any stressful circumstance, whether deliberate or involuntary. With your prefrontal cortex function essentially impaired for the first thirty seconds after you get in the cold, all your thoughts and actions initially Will default to removing the stressor. It's where the immediate "I want out!"

response comes from. The key is to remind yourself that it is normal to feel like that.

You're in what I call "pure impulse" mode. Do this: Try to consciously watch your mind shift from "I want out!" to "I've got this," which Will likely happen again and again very rapidly in that first thirty or so seconds. By observing that phase and not trying to grind your way through it, you are actively engaging your prefrontal cortex in a way that is not typical of someone going through a mild cold-shock response. That's an extremely valuable skill, to be able to observe your own mind during stress. It's not about passively observing or dissociating, nor is it about getting hijacked by stress.

It's about learning adaptive self-control during stressful moments. The first time you try this, it's hard, but most people find that within a few sessions the mind racing starts to lessen, and pretty soon, they feel in control of their thoughts and actions from the start. Few people have mastered this valuable skill in any domain of life, but anyone can learn it. Deliberate cold provides a safe way to practice. I believe the walls approach mentioned above is the best way to practice this skill, whose benefits translate to other forms of stress.

When we measure time elapsed in the cold, we focus, not surprisingly, on its passage. As you learned

in the introduction to this chapter, high arousal states tend to Slow our perception of that. We count, we wonder Why time seems to be moving so slowly—a known function of stress— and we focus on it... 3, 2, 1, Okay, done! Reallife stressors are very different. Some hit when we least expect it (eg., an accident or crisis), others are stressful in large part because we know they are coming (eg., a public talk we need to give, a medical appointment we would sooner avoid, etc.). The way to train for real-life stress in the cold is therefore to learn to sense the increases in adrenaline that occur as we approach, enter, stay in, and exit it.

Each of these can be the walls you need to climb over. If you think like that, you'll notice something very useful: Stress is not constant. It's not, as we say, a step function. It oscillates, often very quickly and irregularly. The walls appear, and then disappear and reappear with unpredictable frequency.

If you dread the cold, remember that adrenaline is released merely from the anticipation of getting in. Congratulations! That's the first wall. And as we lower ourselves into the cold, that's another wall; next comes the first "I want out!" wall, then it passes... then comes the next wall, the next, and the next. Importantly, the time between walls is variable.

This is also important to notice, as that is exactly how reallife stress feels. We often can't predict when the next stressful event is coming. Basically I'm suggesting you approach deliberate cold exposure as resilience training. And by doing it you'll become mentally stronger in the face of nonvoluntary stress. Just be sure you don't stay in the cold longer than five minutes.

Stick to the one to three minutes recommended here, or "go over" two to eight walls. Then get out and go about the rest of your day. Some people, myself included, like to take a hot shower after deliberate cold exposure. People debate whether it's better to end on cold or not. I don't care.

I like a hot shower after and, in my experience, it does not seem to diminish any of the benefits. Find what works for you. As a final point, you actually have more moment-to-moment control over the temperature of the water than you may think. While you can't quickly make

the water temperature change, you can dynamically adjust your experience of it. If you hug your arms around your body, raise your shoulders toward your ears and bring your knees up toward your chest, you Will feel slightly warmer.

Conversely if you keep your arms and legs extended and move them around, you'll break up the thermal layer of heated water around your body, making the experience much colder. Want it even colder? Make sure to let the cold water reach as high as your armpits and keep your hands and feet under. You'll want to remind yourself that this is all very good for you now, and in the hours to come. Cursing Selye, as I do, or cursing me (feel free), is allowed and encouraged.

COMMON QUESTIONS

Is deliberate cold exposure safe? Cold exposure can be dangerous if the water is exceedingly cold. Exercise caution and prudence with this tool. If you are new to cold immersion, start off slowly, with short exposures in

temperatures that are outside your comfort zone but not extremely so. Try a cold shower, which allows you to adjust the temperature higher or lower quickly. Alas, there is not much data on deliberate cold exposure for pregnant women so if you're concerned about potential negative outcomes, avoid this. Don't do it when sick with a cold, the flu, or other respiratory illnesses. Regardless of what you may see online, it's likely to make your illness worse.

Here's a good set of guidelines for nonpregnant individuals with no serious health issues: • Start with water around 50 degrees Fahrenheit the first time, maybe in the midto high 40s. Test the water temperature with your hand or foot before getting in; it should feel cold and make you think, "Whoa, that's cold!" but it should not "burn." • Ifthe water triggers a reflex to raise your shoulders and huddle your arms when you get in, then it's cold enough. Ifyour breathing involuntarily quickens, it's cold enough. • Get in up to your neck with your feet under. Put your hands under last. • Ifyou opt for a cold shower, back into the water until your whole body and head are underneath the stream. • Expect your mind to start racing with thoughts like, "I want out! Why am I doing this?!" Congratulations, you have a functional autonomic nervous system.

Note: Never, everjump or dive into very cold water. Lower yourself in while holding something reliably stationary like a ladder or the sides of a cold plunge, or back into a cold shower. I recommend against

into two to four sessions, they enhanced the amount of brown fat in their bodies, a healthy, mitochondria-rich fat that surrounds the heart, neck, and clavicles.91 That additional brown fat allowed them to feel more comfortable in other cold environments, because when the sympathetic nervous system is active it causes brown fat to burn fatty acids and glucose, which generates heat in the process. In that way, brown fat is like an internal furnace. So if you're someone who is often cold, deliberate cold exposure can help shift your thermal set point. This is connected to the metabolic benefits of cold immersion, which are outside the scope of this protocol. (Note that the participants didn't gain weight; they just added this special—and beneficial—type offat.) I hate the cold so much. Is there any alternative method?

Yes. As an alternative to deliberate cold exposure, you can do three to five minutes of cyclic hyperventilation awayfrom water, which reliably raises adrenaline levels. Cyclic hyperventilation, as the name suggests, involves taking long, vigorous inhales through your

nose and passively exhaling through your mouth—think of it as letting the air fall out rather than forcing it out —over and over. It's not easy to keep up this breathing for several minutes. You'll find it makes you quite agitated and a bit lightheaded.

Data suggest that cyclic hyperventilation works similarly to cold exposure.22 Adrenaline rises. Your heart rate increases. Your limbs Will feel like they want to move. And your brain Will ask "Why am I doing this?!" It's normal to feel some light burning in your nostrils as well.

It feels uncomfortable while you're doing it, but it can improve mood and sleep afterward. Also, you can back off the intensity and/or duration of the inhales and thereby modulate your stress response as you go, an advantage that cold exposure doesn't Offer. So if you really hate the cold or if you don't have access to it, cyclic hyperventilation is the next-best alternative. Just never do it in combination with deliberate cold exposure as, again, you can pass out and drown. People have died combining cyclic hyperventilation with cold immersion; that's Why I'm repeating the warning here yet again.

And of course never do cyclic hyperventilation while driving or operating machinery, or other activities that demand your attention. Just sit or lie down somewhere safe for a few minutes to use this tool.

STRESS CONTROL PROTOCOL Do Non-Sleep Deep Rest (NSDR) to Buffer Against Stress

WHAT TO DO

• Do a non-sleep deep rest (NSDR) session for 10 to 30 minutes. Sit or, ideally, lie down in a quiet place and listen to a scripted NSDR session. Guided NSDRs in different voices are available online (hubermanlabcom/nsdr) for zero cost and can be downloaded to use offline (also at no cost). NSDR is especially useful in highstress times lasting weeks or longer, such as intense work or school periods, or when facing deadlines and life transitions, such as raising young kids or caring for an elder. Ideally you do NSDR once per day but it is beneficial to reduce stress at any point.

I do it at least once a week, depending on my schedule.

HOW IT WORKS

I talked about NSDR earlier in this book (remember Sleep Protocol It is impossible to overstate what an extraordinarily potent tool NSDR can be for mental and physical health. In chapter 1, I covered NSDR's ability to restore mental and physical energy without creating

energy-replenishing effects.28 We also know that people typically emerge from NSDR in an alertbut-calm state of mind and body, that center of the seesaw. When one considers that it costs nothing but ten to thirty minutes of time, that is quite a return on investment! As discussed in chapter 1, NSDR involves shifting one's focus throughout the session, which for some people makes it easier than meditations that demand you anchor your focus or "clear your mind." NSDR asks that you shift from thinking, doing, and planning, to being and feeling—l know, that sounds super New Agey. But the instruction is based in physiology, not mysticism. Here is why: Turning off our thoughts is really hard.

In fact, we can't do it. Even in sleep, we dream. Even in REM sleep people can solve and respond to math problems.22 (If you're wondering how the experimenters knew that, given people are paralyzed during REM, the subjects signaled correct answers to simple subtraction

problems using eye movements.) So forget trying to destress by not thinking. It just won't happen. We can, however, easily shift our immediate experience from planning and analyzing to just experiencing physical sensations.

NSDR has us do that. We deliberately shift our attention from one aspect of our sensations in that moment— for instance from our toes to our fingers, then to a location away from our body such as to sounds in the room, then to sounds far away, then back to a specific location on our body, and so forth. This regular and deliberate shifting of our focus on different aspects of our perception both eliminates the strain of trying to continually focus on one thing, and the tendency of our

mind to default to planning and analyzing. With NSDR, your mind drifts from one sensation to the next in an intentional but relaxed way. When you do that, your mind goes into an atypical, almost liminal state, the one described above— somewhere between that of deep sleep and being awake. Cortisol levels drop, discrete brain areas fall asleep while others remain activem Because NSDR is done lying down with eyes closed, from the outside it appears you are napping but you're not. You're in self-directed reset mode.

Ifyou've never done yoga nidra or NSDR before, this may all sound a bit abstract, but it's really very concrete. It is grounded in what we know about self-directed brain states. And it's very easy to do. One of the biggest advantages of NSDR is that, in contrast to mindfulness meditation— which is a skill you build up gradually over time and that requires daily (or near daily) practice to maintain the positive effects—NSDR can work the first time, and every time. You lie down, listen to the instructions, and voila!

Ten to thirty minutes later you emerge feeling renewed and rested. Most everyone who tries it marvels at the immediate positive effects. It's starting to be used in schools, by elite military operators, professional athletes, Fortune 500 CEOs, and many other men and women around the world as a tool for managing stress and even as a means to accelerate recovery from illness and injury, to offset some of the negative symptoms of menopause, sleep deprivation, trauma, and I encourage everyone to try it a few more. times and experience the benefits for themselves.

COMMON QUESTIONS

When should I do NSDR? This is the most common question I get about NSDR. You can do it ANY time. In the morning if you wake up and don't feel fully rested and don't have time to get another hour or more of sleep. In the afternoon instead of a nap and ifyou know you'll be working harder or longer than usual that day.

Before a workout if you're feeling less-thanmotivated or lacking energy. In the evening ifyou get home and want to destress from the day. In the middle of the night if you wake up and can't fall back asleep. (In that case, get out of bed and do an NSDR then get back into bed; a return to sleep Will arrive easier.) The simple answer I default to whenever asked "When should I do NSDR?" is "First thing in the morning, or in the afternoon, or any time of night.. " It is that versatile and that useful.

STRESS CONTROL PROTOCOL Do a Micro Meditation Immediately After Waking Up

WHAT TO DO

• Incorporate a simple, five-minute daily meditation into your routine. Any time of day can be beneficial, but doing it before getting out of bed each morning ensures you don't have to carve out time later and may Offer distinct benefits as well. • If you feel an urgency to get out of bed before five minutes have elapsed, stay put. Contrary to how it is usually framed, meditation is one of the best forms of resilience training. So see this as an opportunity to practice resilience. This meditation is not about clearing the mind.

It's about building the capacity to sit with the stress that even micro meditation can trigger. This Will help you better deal with real-life stressors once your day starts.

HOW IT WORKS

Plenty of evidence exists that mindfulnessbased meditation practices buffer us against stress, and even brief daily sessions reduce anxiety.— I mention both stress and anxiety

when you wake up and you have to dive into your day. Your mind wants to shift to planning and doing. In that way, meditation can serve as a way to increase your stress threshold. Not unlike learning to stay in uncomfortably cold water. This is as hard as it sounds, but well worth it.

Regular meditation teaches us to observe, rather than act on, our autonomic arousal. It also reduces the likelihood that real-life stressors Will hijack our mind. Just like Stress Control Protocol 2. One study found that the intensity of the emotions people experienced while viewing potentially disturbing pictures was reduced when people did so from a "mindful" state, indicating that meditation promotes a kind of awareness that makes a person's autonomic nervous system less subject to outside stimuli.U06 Indeed, neuroimaging studies have confirmed that meditation consistently alters brain regions connected to both body awareness and mental awareness, as well as emotional regulationl The underlying mechanism appears to be connected to the production of GABA, an inhibitory neurotransmitter—meaning a chemical that generally suppresses the firing of other neurons. People with panic disorders and anxiety tend to have decreased GABA activity in particular regions of the brainl Researchers have theorized that the increased blood flow to the prefrontal cortex that seems to occur during certain types of meditation may, in turn, lead to increased GABA activity, along with improved top-down inhibition of brain areas that generate reflexive thoughts and actions, and this equates to a general sense of relaxation.

A bit more on the format of the micro meditation: stay in bed or, if you must, get up and use the bathroom first, and then set a fiveminute timer, sit or lie down, and stay as Still as you can while you focus on your breathing. Observe where your mind goes. If it wanders— which is completely normal by the way—redirect your focus back to your breath. Slow nasalonly inhales and exhales are best. But the most important thing is to do it, and on days when you feel stressed, do it anyway.

In fact, that's a bonus opportunity. The benefits to your ability to observe your stress rather than have it engulf you that come from a daily five-minute meditation are well worth it. And of course, if you want to do a longer meditation, go for it. As simple as this protocol may sound, I and countless others can attest to the fact that it provides immediate benefits, and benefits that extend throughout the day. And one more thing: We don't yet understand the science of exactly how, but it also seems to create the ability to "see" when stress is coming your way.

Perhaps it sensitizes the brain circuits that respond to autonomic arousal. Perhaps it leads to better recognition of adrenaline in our body rising, or our heart rate starting to increase. After all, a five-minute meditation Will gradually have you feeling more anxious as you progress from start to finish; it's very different from getting into cold water, which causes a big spike in adrenaline right at the start. But that ability to sense approaching stress is a game changer, as it can reduce your reactivity and improve your ability to make decisions about what to do and say—and

in many cases, what not to do and say—when the inevitable stressors of life appear.

STRESS CONTROL PROTOCOL Boost Your Morning Cortisol and Catecholamine Peak

WHAT TO DO

• Stack three to four of the pro-cortisol and procatecholamine triggers below in the morning, and limit them at night. • Get bright light, ideally sunlight, in your eyes for about ten minutes, as soon as possible after waking. If you wake before the sun is up, you'll either have to wait for the sun to rise or use a 10,000 lux light for this trigger. (I cover how in detail in Sleep Protocol 1.) • Hydrate with 16 to 32 ounces of water and consume some caffeine in the first two hours of your day. How much caffeine? 100—400 milligrams depending on your body weight and caffeine tolerance. A strong cup of coffee or yerba maté (my preferred caffeine source) has 150—200 milligrams of caffeine. Note: Chronic caffeine drinkers don't get as much of a cortisol increase from their usual amount of caffeine as those who aren't, but it Still has a positive effect on other catecholamines for reasons covered in chapter 1.110

You may recall from those protocols that deliberate cold exposure doesn't spike cortisol all that much, but it does significantly increase the other catecholamines. And that is beneficial here. • Consider occasional (once or twice per week) use of low-dose licorice root containing glycyrrhizin or consume grapefruit in the first one to three hours of your day. Both have been shown to significantly increase cortisol and the duration of cortisol action. Subjectively, they can increase feelings of morning and daytime energy and focus. Warning: Despite licorice and grapefruit sounding natural (and they are), their effects can be very potent.

They can increase blood pressure and/or fluid retention at high doses, so use the minimal effective dose and, as with all supplements discussed in this book, consult your doctor first.

HOW IT WORKS

Cortisol and the catecholamines adrenaline and norepinephrine, but especially cortisol, are often thought of purely as stress chemicals. The third catecholamine—dopamine—escaped this reputation but is usually thought of (also incorrectly) as purely a pleasure chemical. And while cortisol is involved in the stress response, thinking about it as purely a stress chemical is wrong. I wish I could banner this across the sky: Cortisol Is Not a Stress Hormone! In fact, if I could impart just one message about how to improve your health, it would be this: Strive to get your morning cortisol high (yes, high!) and your nighttime cortisol low.

The main function of cortisol is to ensure there is enough energy in the form of glucose, or blood sugar, for you to meet life's demands. Yes, that can include overcoming stressors, but it also includes simply waking up each morning with the physical and mental energy required for work, school, family, and exercise. Cortisol causes glucose (as well as fats) to get released into the bloodstream, and it also stimulates the liver to make more glucoseun Cortisol and

adrenaline play additional key roles in our body's everyday functioning. They improve our immune system's ability to ward off infections, they elevate our mood, they allow for our capacity to form new memories, and they increase metabolism... so just forget the idea that cortisol and adrenaline are only about stress. A gradual rise in your cortisol levels starts three to four hours before you wake up each morning and its levels eventually hit a threshold level, at which point you wake up, open your eyes, and get out of bed.

Keep in mind that cortisol doesn't act alone to rouse you from sleep; a parallel rise in body temperature and clearance of the sleepiness-inducing molecule adenosine follows a similar schedule. Nonetheless, scientists call this rise the "cortisol awakening response" or CAR. Now that makes it sound like a medical problem of some sort, but CAR is a normal and (provided it is timed correctly) health-promoting aspect of our biologyu-Z And it helps buffer you against unhealthy stress. Cortisol first starts to rise as a result of a signal from your SCN, your brain's master circadian clock that we discussed in chapter 1. For most people, that rise starts somewhere around

five hours into their night's sleep and continues sharply over the subsequent two to three hours until it reaches a threshold level and bam! You wake up. Even if you use an alarm clock, you can be sure your cortisol levels were rising as you approached the start of your day. Now, once you're awake, cortisol continues to increase until it hits its peak for that twenty-four-hour cycle. For most people, that peak occurs anywhere from 0.5 to 1.5 hours after waking up.

Then cortisol levels slowly start to decline— a process that continues into the late afternoon and evening. Assuming you encounter no major stressors, your cortisol levels continue to drop in the hours before you fall asleep and reach their lowest level about three to four hours after you doze off, and then slowly start to rise again to eventually trigger the next morning's CAR, and so on. This is the ideal cortisol pattern for mental and physical health, energy, and resilience to stress. Night shift workers deal with a very disrupted sleep-wake schedule, and their cortisol pattern gets fragmented, with multiple peaks and troughs throughout the day. For some, the pattern inverts to match their wake-sleep schedule.

Although that might seem like not that big of a deal if they are Still getting enough sleep, it's known to cause serious and longlasting health issues. Indeed, many of the known health detriments of nocturnal schedules are due primarily to alterations in the pattern of cortisol release and its downstream effects. I cover what shift workers can do to partially offset those negative effects in chapter 1 and chapter 5.113

depression, and weight gain can result, as can physical imbalance, leaving a person feeling unsteady on their feetu-6 Studies have also linked abnormally high cortisol levels later in the day and into the night to acne, irregular menstrual cycles, and digestion issuesll High cortisol at night has been linked to insomnia and reduced amounts of REM sleepll Given this, everyone would do well to try to anchor and amplify their cortisol and catecholamine peak in the first hours after waking to start their day and limit their cortisol and catecholamine levels at night. That simple framework goes a long way toward limiting your chronic stress. And that's what this protocol is all about. Let's start with what to do at night, because how you prepare for bed affects your cortisol and catecholamine levels the next day. Some of what you're about to read Will be familiar to you from chapter 1.

Before bed, implement a wind-down routine of dimming the lights, maybe taking a hot shower or bath or even washing your face with hot water. If you're feeling stressed from the day, take a few moments to do some physiological sighs to lower your heart rate, and/or do a tento

thirty-minute NSDR session. Don't drink too many fluids or eat in the two to three hours before sleep, and definitely try to avoid caffeine in the eight to ten hours prior to bedtime. Viewing bright overhead LED lights (of the sort that are in many bathrooms) is a quick way to spike your cortisol and quash your melatonin, and checking your email or the news in the last hour before sleep is definitely a risky move if you want to keep your cortisol low at night (see chapter 1 for

details). And to be clear: Low cortisol at night should be your goal! If you happen to wake up in the middle of the night for whatever reason, avoid viewing bright lights or looking at your phone. Again, do some physiological sighs to help keep your heart rate low, and let yourself drift back to sleep. If you can't fall back asleep, listen to an NSDR session.

Or try the weird eye-movement trick I shared in chapter 1. By the way, if you exercise in the evening or at night, that does significantly increase lateday cortisol. It's better to exercise than not to, if you have a workout planned. To help keep your cortisol and catecholamine rhythms on a diurnal schedule, you can reduce the intensity of your evening training a bit and/or be sure to include a wind-down routine afterward. Andy Galpin, an exercise physiologist, encourages this after all workouts, but especially nighttime ones.

Even moderate-intensity exercise can cause a large increase in cortisol (as discussed in chapter 2), especially if you consume caffeine firstu-2 That is one more reason to drink caffeine before moderate to intense exercise in the morning, but also to avoid it before moderate to intense exercise at night. I don't want to demonize late-day exercise. Better to do it than not do it. But just know that if you exercise intensely within four to Six hours of bedtime, your next-day waking cortisol may be lower due to the negative feedback regulation of cortisol.U2-Q That's a mechanism you often see with hormones in the body, like cortisol or testosterone: When those hormones reach a certain high level, it signals specific brain

areas and glands, such as the hypothalamus and pituitary and adrenals, to reduce their release of those hormones. Just another reason Why spiking your morning cortisol is a good thing. It not only sets you up for the day, it helps keep cortisol levels low at night.J-Z-l So if you exercise at night and wonder Why you're dragging a bit the next morning, this may very well be why. Be diligent about using the tools in this protocol and you'll be A-OK. What about spiking your cortisol and catecholamines in the morning—how best to do that?

There too you have options: Viewing bright light can boost your morning cortisol peak by 50 percent, making it a very easy, effective and beneficial practice (see Sleep Protocol 1 for more). Sunlight is ideal, but a 10,000 lux light works too.J22 Hydration, caffeine, moderate to intense exercise, deliberate cold exposure, maybe cyclic hyperventilation (but never with deliberate cold exposure): Doing one of these is better than none, two is better than one, and so on. If I had to place them in order of effectiveness, I would say: sunlight viewing, exercise, and caffeine are the big levers here, but don't worry about the order. Let logic be your guide. Obviously hydrate before and during your exercise, drink caffeine for an energy boost, get the sunlight before or during exercise if you can, but if the sun isn't up and you're ready to train, train!

Don't wait for the sun to rise, thinking that it has to come first because it's a big lever. The cortisol and catecholamine systems are primed for release in the early hours of your day and Will synergize without the need to get the relative timing (of all the things that cause their release) perfect.

metabolism of cortisol, the net effect of which is to increase cortisol's duration of action. Licorice root extract contains something called glycyrrhizin (pronounced gly-cer-ry-zin) that potently inhibits the same abovementioned enzyme 11 P-HSD2. You may be thinking, Grapefruit and licorice?! Is he serious?! Yes, quite serious.

There are human data to support the use of each to increase cortisol's duration of action including studies demonstrating dose-dependent effects. So much so that well-informed MDs Will warn patients with Cushing's syndrome—which is a condition marked by pathologically elevated levels of cortisol—to avoid grapefruit, licorice root extracts, teas containing glycyrrhizin, and any licorice candy. (Note: Some traditional black licorice root candy, typically sold in Scandinavian countries but elsewhere too, contains glycyrrhizin, though versions without glycyrrhizin exist too.) I suggest most everyone avoid licorice root extract consumption late in the day and at night. If you want to experiment with taking licorice root extract that contains glycyrrhizin please talk to your doctor first. At high doses it can raise blood pressure, cause fluid retention, and even affect the metabolism of certain pharmaceutical drugs and potassium because of the many roles of the abovementioned enzymes in the liver. I use licorice root extract with glycyrrhizin as a tool occasionally (once or twice a month).

I find that 100 to 250 milligrams of licorice root in capsule form is enough to feel its effects. I make sure to use a version containing glycyrrhizin, and I take it with my morning caffeine usually before

a hard workout that I need to start earlier than is typical for me. The energy-increasing effect is very noticeable. On very rare occasions I Will take it around noon to try to extend my window of elevated cortisol to work for an especially long day, or to exercise hard midday. Again, I use it as a sometimes tool, not a regular tool, to boost cortisol in the morning. Overall, grapefruit and grapefruit juice have a milder, but Still meaningful, impact on cortisol's duration of action.

Most people can consume these without an adverse effect. But they can interact with various medications, so be sure to consult your doctor before taking. If you'd like to try incorporating grapefruit as a cortisolboosting tool in your routine, perhaps eat a grapefruit or drink a small glass of grapefruit juice in the morning (Pink or conventional is fine; it doesn't matter). See how that affects your energy throughout the day. I would avoid these at night for obvious reasons.

One more thing to note here is that it is hard to find very-low-dose licorice root extract with glycyrrhizin. Most are 400 milligrams or more, and even at 200 milligrams it works very well to increase energy, so I empty half a 400 milligrams capsule into water or my coffee, and drink that. It tastes good. Remember: It raises blood pressure, and taking glycyrrhizin every day can lead to serious problems over time, so if you experiment with it, try a low dose—even if it means having to open up a capsule to remove a portion of the contents. Once more, talk to your doctor before you do.

And ifyou have high blood pressure, don't take it!

I can't end this protocol without one more reminder about caffeine: If you consume caffeine in the late afternoon, too, don't overdo it. I get it; I love yerba maté in the morning, and I love, love, love the taste of coffee in the afternoon. But if you're an afternoon caffeine drinker and having sleep issues or trouble getting going in the mornings, you can bet it's at least partially

because the caffeine levels in your bloodstream are Still too high from your afternoon coffee or tea. Try to drink only decaf after 2:00 pm., or limit yourself to roughly 60 milligrams of caffeine (that's about 3 to 4 ounces of regular coffee, or an espresso shot), and do your best to avoid it altogether within eight hours of bed (refer to Sleep Protocol 11 for more guidance). You could also opt to ditch the afternoon caffeine and instead to take 300 to 900 milligrams of alpha-GPC, which is a short-acting stimulant that can increase energy and focus and may actually improve REM sleep at night if taken Six to ten hours before bedtime.

I discuss this tool more in chapter 6. This protocol includes several tools, so to recap: By increasing the slope and height of your morning cortisol and catecholamine peak, you buffer yourself against later-day stress in several ways. First, you're likely to experience improved energy and focus for many hours afterward. Viewing bright light,J24 consuming exercise, and cold exposureU2-Z all increase cortisol and/or catecholamine release. This helps us keep our reactions proportional to any stressors we may encounter later.

By starting your day in a state of focus and readiness, if a stressful situation arises you don't need to

STRESS CONTROL PROTOCOL Consider Specific Supplements to Help Offset Excessive Stress

WHAT TO DO

• Find a combination of supplements that might help you combat severe or prolonged periods of stress. Remember: Behavioral protocols are the foundation of mental and physical health and performance, and that is especially true when it comes to stress control. It's also true that ifyour sleep duration and quality are not suficient, no amount of pharmacology Will help, at least not for long. Refer to chapter 1 for more on that topic. With that said, below is a list of supplements that have shown some eficacy for reducing various aspects of the stress response: • Ashwagandha: 200—300 mg, 1 to 2 times per 24 hours, but taken only in the afternoon and/or evening. • L-theanine: 100 mg or 200 mg, 1 to 3 times per day; if you drink caffeine regularly or are in a period of unusually high stress, take theanine with your caffeine. • Phosphatidylserine: 400—800 mg, once per day anytime after 4:00 pm.

well aware that some people Will opt to just dive in and take all four because they hate stress that much. Try to hold back.) Ashwagandha: Ashwagandha, also known as Indian ginseng or winter cherry, is used as an adaptogen— a term that broadly refers to substances that improve a person's resistance to stress. A large review published in 2021 that included fortyone total studies in humans included seven that focused on stress and anxiety, enrolling a total of 491 adults who either reported being under high stress or were diagnosed with an anxiety disorder. I Ashwagandha dosages varied from 240 milligrams per day to 1,250 milligrams per day for an extract and up to 6,000 milligrams when administered as a powdered root. (Most ashwagandha capsules sold over the counter in the US contain 200—600 mg per capsule.) Taken together, these studies showed that ashwagandha reduced stress and anxiety when compared to a placebo, and also reduced sleeplessness and fatigue. Cortisol levels in the blood were also lower among people taking this supplement, and it appeared that the cortisol-lowering effect of ashwagandha is likely the major basis of its effect on lowering stress and anxiety.

A study of Sixty adults found that those who took 225 milligrams or 400 milligrams per day of ashwagandha reported reductions in anxiety, depression, and food cravings compared to

people randomized to a placebom One study found that men and women taking ashwagandha had a 22 percent and 25 percent reduction in cortisol, respectively,

when they took ashwagandha for Sixty days versus taking a placebo for Sixty days.J-32 Research has identified several components that could contribute to the stress-reducing effects of ashwagandha. Most likely the effect is due to a "cocktail" of several different compounds in ashwagandha that work in parallel on various aspects of the hypothalamic-pituitaryadrenal axis, which, as you may remember from Sleep Protocol 1, is our main biological circuitry for mobilizing a physiological response to a stressor. The most potent of these mechanisms is the suppression of the signaling pathway for corticotropin (ACTH), a hormone that acts to stimulate the release of cortisol from your adrenals.J3.3 The ashwagandha-suppression of corticotropin (ACTH) happens through a multistep process that begins with compounds called withanolides (naturally occurring plant hormones). These compounds can lower corticotropin-releasing hormone (CRH) in the hypothalamus.J-34 Limiting the amount of CRH in turn dampens the signaling that would normally trigger the release of cortisol. The fact that ashwagandha works "upstream" from the adrenal glands has a potential downside: Namely, it is likely to impact all the biology connected to CRH, which includes the gut and the immune system, among others.

That's one of several reasons Why I suggest not taking ashwagandha at dosages higher than 600 milligrams per day for more than a few weeks before taking at least a few weeks off. There is at least one report of someone taking a high dose (950 mg) of ashwagandha daily for a year and developing adrenal insuficiency. People seem to vary quite a bit in their sensitivity

to ashwagandha, so if you decide to try it in times of excess stress, I suggest starting with one dose per day, taken in the evening or an hour before bedtime at a dose of about 100 to 200 milligrams, and only increase the dosage, if needed, to find the minimal effective dose for you. Importantly, you don't want to suppress cortisol in the early part of your day. If you decide to take more, take ashwagandha in the afternoon around 3:00 to 4:00 pm., and perhaps again an hour before bedtime in a dosage of 200 to 300 milligrams each. As mentioned above, I suggest you do not take ashwagandha at doses exceeding 600 milligrams per day for periods of longer than a couple of weeks. While the reports are largely anecdotal, some people taking high doses of ashwagandha (700—1,200 mg) per day experience apathy, starting about three to four weeks after continuous intake.

The reasons for this are not Clear and may be similar to the case report mentioned above, but because eficacy has been shown at lower doses of 300 to 600 milligrams per day, I suggest people use it as a tool to augment stress reduction by taking about 300 milligrams on a regular basis as described above, or reserve its use to high-stress periods and take higher dosages (greater than 600 mg/day) only during those times. And don't let up on the behavioral practices for stress reduction. A quick note about sourcing ashwagandha and other herbal adaptogens: The herbal supplement world is full of fake and contaminated products, perhaps even more so than for other types of supplements. Stick to major brands with thirdparty testing, especially ones that test for and

a placebo for four weeks.L4-Q The investigators, a team from Japan, used several standard measures to gauge depression, anxiety, and sleep quality. They reported that the group taking L-theanine had significantly fewer stress-related symptoms. In another study, a group of researchers in Australia and the Netherlands enrolled thirtyfour adults between the ages of eighteen and forty in a blinded study to measure stress response in reaction to a series of

cognitive tests involving mental arithmetic, memory, and coordination. 141 Tasks like these can trigger stress because they challenge us. The tasks were presented following consumption of either a placebo or L-theanine as a nutrient drink. They found that the stress response, measured through the amount of salivary cortisol, was reduced compared to the placebo, an effect that appeared to last for several hours.

In addition, the L-theanine group showed an increase in alpha brain waves, suggesting a more relaxed state of mind. That being said, cognitive performance was no different between the L-theanine and placebo groups. The reduced stress reaction may be beneficial for other reasons, but it's useful to know that theanine didn't make people better at the tasks they were given. In terms of the underlying mechanisms, L-theanine research has found that this compound indirectly increases GABA—the major inhibitory neurotransmitter in the brain—and dopamine.1 2 Theanine taken in the hour before bedtime is part of the sleep Stack I describe in chapter 1. There, I warned that people who have especially vivid dreams not take theanine.

I suggest that

people who want to try theanine to reduce their stress do so by taking about 200 milligrams of it in the early afternoon and perhaps another 100 to 200 milligrams again in the evening, especially if you had caffeine in the early afternoon. Phosphatidylserine: Phosphatidylserine is a type of phospholipid. It is naturally found in the membranes of neurons, and comprises about 10 to 20 percent of the total phospholipid content of the brain—an enormous percentage when one considers that much of the nonliquid content of the brain is phospholipids. Naturally occurring phosphatidylserine plays a role in cognitive function,J-43 and the loss of phosphatidylserine in neurons known to occur in later adulthood may partially explain age-related (non-dementia) cognitive decline. What's more, in its natural state in our brains, this compound is known to decrease the stress response.U44 As a supplement, phosphatidylserine has shown some benefit in Older adults who experience chronic stress.

There is some evidence phosphatidylserine taken late in the day may preferentially reduce evening and nighttime cortisol,U4.5 which may explain how it indirectly supports sleep. In one investigation, from 2004, participants took a phosphatidylserine supplement at a dose of 400 milligrams, 600 milligrams, or 800 milligrams, or a placebo (twenty people per group) and then completed a type of social stress test that involves performing in front of an audience.J-46 Data collected on biological indicators of the stress response revealed a significant decrease in these symptoms among

the 400-milligram group, but not among those taking the higher doses (or the placebo). It's hard to interpret the effect of higher doses, but this is not uncommon in pharmacology studies. Many studies of stress and cognitive performance show an inverted U-shaped response, where lowand high-dose conditions yield similar or no effects, but mid-range dosages do. Such are the complexities of biology. Another study found no link between phosphatidylserine and measures of a stress response across all Sixty participants, but the researchers did observe an effect among participants with high chronic stress levelsl Initially, this group had a diminished cortisol rise in response to a social stress test.

That may sound like a good thing but it's not. A blunted cortisol response is often a sign of underlying issues, such as depressionl Treatment with phosphatidylserine appeared to restore the healthy rise in cortisol in response to the test, meaning it seemed to recalibrate the stress response rather than stop it. I say all this not to confuse you but to remind you that though the best model for your autonomic nervous system might be a seesaw, we need to consider the

"hinge" of that seesaw (metaphorically speaking). How it bends, or Sticks, matters. And compounds like phosphatidylserine may change people's stress response by adjusting mechanisms analogous to how you might treat a hinge, rather than merely increasing or decreasing cortisol or adrenaline.

Finally, in a six-week study from 2014 of seventy-five men randomized to phosphatidylserine (at a dose of 200 or 400 mg per day) or a placebo, the higher dose of the

supplement appeared to lower the stress response among participants with chronic high stress. 1 If you opt to try phosphatidylserine for stress control, I recommend starting with a low dose of 100 to 200 milligrams, taken a few hours before bedtime, and increasing the dose only as needed. I know many people who like its effects on lowering stress late in the day, and for sleep. Rhodiola rosea: Rhodiola rosea is a yellow-flowered plant that grows in northern regions of Europe and Asia and on the Atlantic coast of North America. It's long been part of traditional medicine practices in Europe and Asia, used to address fatigue, mild anxiety, and symptoms of burnout.U5-Q It is considered an adaptogen, because it alters the subjective experience of effort, apparently by adjusting the threshold for perceived effort, and by increasing effort endurance in physical and mental exertion. Subjectively, people taking Rhodiola rosea report less post-physical-exertionrelated fatigue and that mental stressors feel less stressful, meaning they can lean into sustained effort without feeling exhausted.

Indeed, Rhodiola rosea has been shown to reduce the symptoms of burnout, which is a multidimensional feature of chronic stress that includes the "wired but tired" feeling that is a common consequence of prolonged periods (days, weeks, etc.) of stress and altered sleep-wake cycles. European medical authorities approved the use of Rhodiola rosea for short-term relief of stress in 2012&5.1 and data on this connection have been slowly accruing.

cells make in response to stressful conditions such as the excessive heat or cold, the presence of heavy metals, or an infection. Researchers are Still investigating the exact mechanisms tied to the stress-reducing effects, as well as its apparent contribution to physical strength and endurancel The picture emerging is that Rhodiola rosea buffers against stress by calibrating the amount of adrenaline that is released in moments of high physical exertion and high emotional stress. I recommend one of two approaches to trying Rhodiola rosea. The first is to take it thirty to Sixty minutes before a known stressor, like a hard workout or big exam or study session or public speaking engagement—basically any event that you know Will require an intense and prolonged effort. I take 120 to 240 milligrams of Rhodiola rosea before hard resistance training workouts, which is per the standard protocol used in most peer-reviewed studies of Rhodiola rosea for physical exertion.

The second way to take Rhodiola rosea is during periods of chronic stress, when you can't get as much sleep as you would like and/or when your output is greater than usual. Think final exam week, the days of long work hours before a vacation, the vacation with your in-laws where you are coordinating everything, et cetera. In those times, I suggest taking 100 to 120 milligrams (the typical capsule dose of Rhodiola rosea in most supplements) three times daily (morning, afternoon, and evening). It can be taken with or without food. I would not suggest taking Rhodiola rosea multiple times throughout the day for more than three to four weeks without taking a break of at least two

weeks, as all the biological pathways Rhodiola rosea is known to impact are subject to feedback loops and downregulation. Use 100-milligramsplus daily dosages of Rhodiola rosea as a shortterm tool to combat stress and burnout, not as a mainstay of your supplement

regimen. And I know I sound like a broken record, but don't just pile on supplements for managing stress. You need—and now have—multiple behavioral tools to reduce stress. Use them!